



OPEN Robust and secure image steganography with recurrent neural network and fuzzy logic integration

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In the time when the communication mode is changing towards the digital aspect, such as; computing devices and the internet, it is natural that there is a need to safeguard the safety as well as the privacy of the data being shared, which can also compromise the integrity. Image steganography is a technique that conceals messages in pictures taken on a digital camera, and serves as a means of secure data trans-mission. In this study, we provide a novel method for image watermarks which is highly secure and robust through the application of Recurrent Neural Systems and fuzzy logic. By merging fuzzy logic with the flexibility of Recurrent Neural networks, the proposed system enhances the protection and rate of hidden information even under noise, compression and other attacks. The model utilizes the learning capabilities of Recurrent Neural networks for optimization of the embedding process, while the Recurrent Neural model is useful for managing uncertainty and improving decision-making processes. Such two-layered systems ensure the inaudibility and invisibility of steganographic data while limiting the embedding of large amounts of visible data, which is highly effective. Robustness and security of the method is tested by using various images types within different environments. To enhance the efficiency and robustness of steganography of images, this work proposes an innovative two-layer method that combines fuzzy logic with Recurrent Neural Networks (RNNs). By enhancing embedding, strengthening attack resistance, and offering an adaptable, intelligent framework for safe data concealing, it advances the field. It was observed that compared to other techniques used in steganography approaches, it is pleasing to note that the whole embedding efficiency and resistance to attacks are far and above enhancement candidates. The results demonstrate that the seamless integration of fuzzy logic and Recurrent Neural Systems provides an efficient and scalable framework for secure image steganography.

Keywords Steganography, Recurrent neural networks, Fuzzy logic, Data security, Robustness

The immense growth of multimedia information over the World Wide Web has created important concerns about information storage and privacy in the current digital era¹. Imaging steganography is growing as an effective strategy to covert communication, in which secret information is hidden inside an image in a manner that makes its existence undetectable². In the presence of difficulties such as noise, contraction, and vicious attacks, conventional steganographic styles have continued problem with keeping a balance between the overall quality of the stego-image, security, and reliability³. Innovative techniques which can improve the flexibility and concealment of inactive data are essential due to these limits⁴. The integration of machine learning and fuzzy logic, particularly through a heterogeneous Recurrent Neural System (RNN) combined with fuzzy logic techniques, presents an effective solution for addressing these challenges⁵. The RNN gives a sophisticated and flexible architecture for data insertion and establishment through the use of the logic power of fuzzy sense and the computational capabilities of artificial Recurrent Neural networks⁵. This approach is extremely resistant to multicolored processing pictures attack as it can efficiently handle nonlinear properties and concerns over the integration process⁶. Contrary to this, fuzzy logic includes mortal-like thinking by handling vague or squishy data, which benefits making decisions while information embedding¹². By combining inaccurate sense with the hardness of Recurrent Neural networks, the steganographic technique is capable of strongly adjust the method of embedding to its features of the covering image, which guarantees perfect data hiding and retrieval^{7,8}. A robust

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and safe photographic steganography method that preserves an excellent degree of visibility, safety, and immunity to attacks is the objective of the proposed method⁹. The system can accurately recognize most suitable areas for data embedding, adapt to differing image circumstances, and safeguard the located data from unauthorized delivery through the use of the Recurrent Neural Systems and fuzzy sensing merging¹⁰. If individuals attempt to take the information, they're unable to since the pieces of information are kept in various locations and it is hidden. Following the obtain, the procedure changes to work, in order to get the original content.¹¹ This study examines the usage of fuzzy sense and RNN in the setting of image steganography, showing the significant enhancement in the safety and durability of the steganographic procedure which may arise from the use of this combination¹². The results suggest that this system excels traditional techniques by means of data concealment, flexibility against attacks, and firmness, leaving it a perfect choice for safe data transfer in an internet location that is quickly growing more secure¹³. In the end, the proposed method presents creative methods that increase the safety and visual appearance of stenography, thus solving possible drawbacks in traditional methods.

Motivation and contribution

Image steganography is a method that attempts to hide private information inside images for safe communication; however, traditional methods address problems including vulnerability to attacks, limited embedding capacity, and vulnerability to image modifications like compression and noise. The motivation for which include Recurrent Neural Systems (RNN) and Fuzzy Logic in steganography comes from the need of addressing these challenges: Vulnerability to Attacks. RNN develops complex patterns while dynamically altering the procedure for embedding to improve robustness to these attacks. Increased Protection using Fuzzy Logic Improved Embedding capacity: RNN will optimize particular parts of the picture to accommodate more storage of information while compromising visual performance, hence improving information-hiding performance while maintaining image authenticity.

This paper's important findings are outlined as follows.

- The protection aspect of the steganographic look at is examined from an alternative perspective.
- A novel method has been proposed which includes the least significant bits, pixel value inconsistencies, and modifies direction exploits using a mixture of ambiguous artificial Recurrent Neural network.
- Encryption of images using LSB, PVD, and EMD methods has been applied to improve the security of personal data.
- Additionally, a type of Recurrent Neural Systems system is used to produce excellent stego images.
- The assessment used various secrets and cover pictures, making sure that the resulting stegoimage demonstrated low noise or data leakage.

The structure of the paper is as follows. Section "[Related work](#)" is a literature evaluation which addresses important issues in steganography system Section "[Recommended system using recurrent neural network](#)" covers an in-depth analysis of the foundations of mathematics and structure of the FNNs. Section "[PCA, EMD, PVD Algorithm](#)" explores the proposed approach, resources, and methods to be used for deep learning Recurrent Neural Systems. Section "[Recurrent neural network](#)" provides findings and analysis in multiple cover and secret images. Finally, the article is concluded in "[Conclusion](#)".

Related work

The deep neural network (DNN) is an instance of artificial neural network which has many hidden units between the data and layers of output. In the identical way, deep categories are tough to figure out and use technology for operation. As Well As it's not only the fact that there are additional axons and connections among them as well. Certain neurons might alter the environment by beginning things up. Comparing commitment to deeper neural networks is demonstrated by the concept of "pre-preparing," that involves establishing the fundamental demands in a neural construction. When start making every load tiny, it takes always to reach to a position in which a thing fascinating appears. The context of a similar way, employing large beginning amounts make one easier to get stopped rapidly in the surrounding optimal. The reason does certain elements of a building structure function effectively while people fail to function well. What types of modifications might be made to make the indicate better this unidentified problem was initially called the "essential credit job matter." The objective starts up a neural network (NN) algorithm or perhaps valuable parts of typical NNs¹⁴. The suggested approach has been compared with previous approaches in Table 1. In this article we utilized the latest method suggested which used PVD and EMD in conjunction with LSB substitution. Table 1 demonstrates how the suggested strategy outperforms the current one with a good margin and a lower mean square error. The neural networks are typically employed to carry out rules according to closure presumption, as either individual investigations or established backslides¹⁵. NNs may undoubtedly outpace additional famed decision-making procedures when the prospect of getting the information grows fuzzy, because of to both the multidimensional structure and the non-linear of the datasets¹⁶. Backward propagation networks (BPNs) have been developed with the ability for creating adaptable links among the original sources of information and the results. A self-decisive continuous limitation may be precisely evaluated by a feed-back NN which involves someone's performing work within the yielding block¹⁷ and established boundaries within the overlaid covering¹⁸. Still, the BPN possesses many kinds of constraints, including the requirement of numerous regulating limitations, the challenge of developing a uniform structure, and the possibility of over fitting the model¹⁹. as an outcome of all these problems, a variety of methods were designed to replace the BPN as a concept²⁰. Still, the difficulties linked to NNs can be difficult in regards to getting ready as well as the possibility to recognize community most effective structures, since the hidden covers and established limitations of a neural network look like a finding²¹. The "fuzzy logic" theory, originally initiated through²², is being demonstrated to have been helpful in handling of problematic as well as

Cover image		Lion			Elephant			Zebra			Lincoln			Deer		
Hidden image		Waterfalls	Mountain	Sea	Waterfalls	Mountain	Sea	Waterfalls	Mountain	Sea	Waterfalls	Mountain	Sea	Waterfalls	Mountain	Sea
		PSNR 48.13233	48.13233	47.03605	45.58863	48.55572	47.44866	45.9962	44.04289	42.96453	41.49364	48.43866	47.37839	45.91944	47.85967	46.78156
Vernam algorithm	MSE	0.999647	0.999647	1.286692	1.795629	0.906793	1.170075	1.634781	2.563248	3.285691	4.610154	0.931567	1.189162	1.663932	1.064419	1.364343
	PSNR	51.6376	51.6376	50.29061	49.41911	51.86575	50.51877	49.64727	54.08495	52.73796	51.86646	48.81556	47.46858	46.59707	55.96796	54.62097
Pradhan algorithm	MSE	0.445985	0.445985	0.608162	0.743309	0.42316	0.577037	0.705267	0.253855	0.346166	0.423092	0.854131	1.164724	1.423551	0.164545	0.224379
	PSNR	61.9649	61.9649	60.61791	59.74641	63.38081	62.03383	61.16233	60.40964	59.06265	58.19115	53.755439	52.40845	51.53695	59.432	58.08501
HFNN algorithm	MSE	0.041361	0.041361	0.056401	0.068935	0.029854	0.04071	0.049756	0.059172	0.080689	0.09862	0.273865	0.373452	0.456442	0.074111	0.10106
	PSNR	63.6789	61.9649	63.23456	62.74741	65.67821	64.54362	63.12909	63.40964	62.906265	60.78115	55.78989	57.40845	54.53695	61.6432	60.56101
Proposed algorithm	MSE	0.028161	0.028161	0.043201	0.055735	0.016654	0.02751	0.036556	0.045972	0.067489	0.08542	0.260665	0.360252	0.443242	0.060911	0.08786

Table 1. Stegno image analysis of proposed and existing system.

unclear datasets. Fuzzy logical thinking is an approach for model-based decision-making which can be used in supervisor structures and interface management to imitate the uniform mental processes for individuals²³. The stegno image using the 1-LSB method²⁴ is compared with the proposed model. Convolutional Neural Network-based image steganography method that operates on DICOM images which conceals the sensitive medical data in the carrier file, thereby increasing the privacy of the sensitive data²⁵. In steganography original message is encrypted by Fractal Sierpinski Gasket (FSG) cryptographic algorithm, that result is hidden into an image using penultimate and least significant bit (PLSB) embedding method²⁶. The Present issues in watermarking medical data include ensuring the watermark's robustness against attacks while maintaining data integrity and confidentiality. The authentic data and information of ROI of a medical image embedded in penultimate and least significant bits PLSB of squint pixels of RONI²⁷. A hash-based approach using the Discrete Cosine Integer Wavelet Transform (DCIWT) to provide secure watermarking for confidential medical data.²⁸.

Recommended system using recurrent neural network

Figure 1 shows the block diagram of steganography. This diagram illustrates a steganography process where a secret image is hidden within a cover image through an embedding process, producing stegno data. The stegno data is transmitted via a communication channel to the receiver. At the receiver's end, an extracting process retrieves the secret image from the stegno data. This ensures secure and hidden communication of sensitive information. Figure 2 illustrates improving visualization and retrieving secret images using RNN with techniques such as PCA + EMD or PCA + PVD. During the integration stage, we divide the image into 5×5 processed data blocks and analyze them in a column-wise manner.

We put these small pieces through an RNN to enhance the image's clarity using procedures like elimination or refinement. To find a hidden picture, PCA + EMD (Principal Component Analysis with Empirical Mode Decomposition) or PCA + PVD (Principal Component Analysis with Pixel Value Differencing) looks at the image and divides it up while the extraction phase is going on. A total of four different kinds of photos are utilized in this technique: the original picture, which is also called the cover image; the secret image, which holds hidden data; the segno image, which includes the secret picture in the covering picture; and the received image, which is used to extract the secret image. The above method systematically performs both the enhancement and the acquisition of hidden images.

The process starts by selecting both the hidden picture and the cover image. The images are divided column-wise into 5×5 blocks, and the equation $\text{Diff} = (1/3) \sum_{i=1}^3 |B_c - B_i|$ is used to calculate the dimension difference. If the variance exceeds 10, PCA + EMD is employed for embedding; otherwise, PCA + PVD is used. A smaller stego figure is then generated, as illustrated in Fig. 3. RNN is utilized to enhance the stego image by

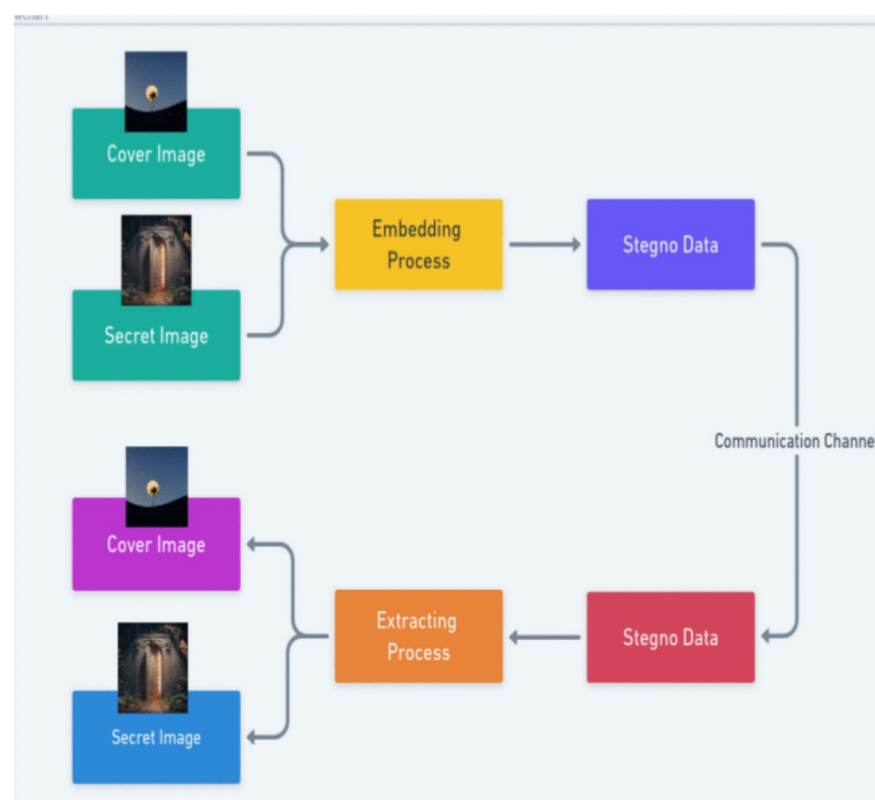


Fig. 1. Block diagram of steganography.

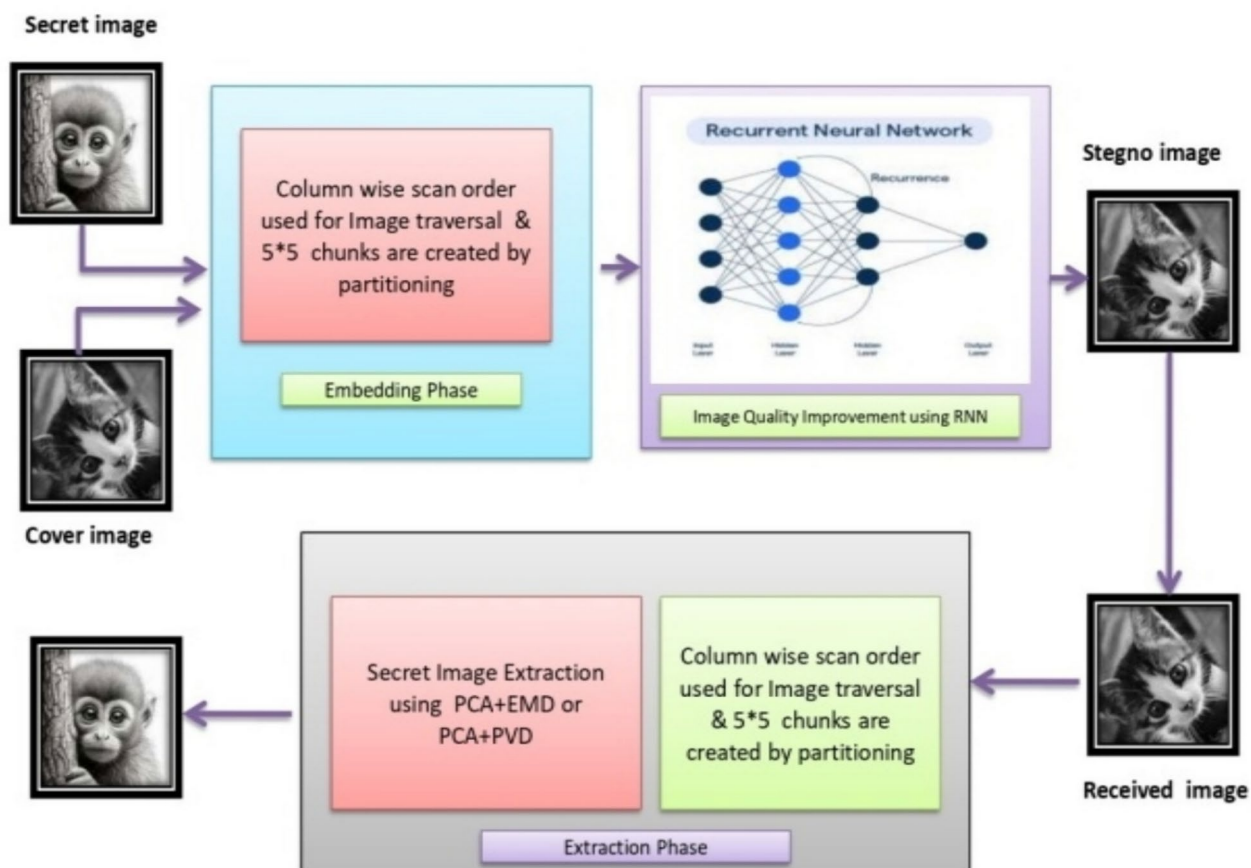


Fig. 2. Block diagram for proposed steganography method.

analyzing and modifying stego attributes and secret bits, followed by a comparison of the improved elements with the cover image. The modified parameters are updated, and if discrepancies remain, backward propagation is applied to further refine the RNN. Finally, the technique shown in Fig. 4 is used to extract the private information, completing the process.

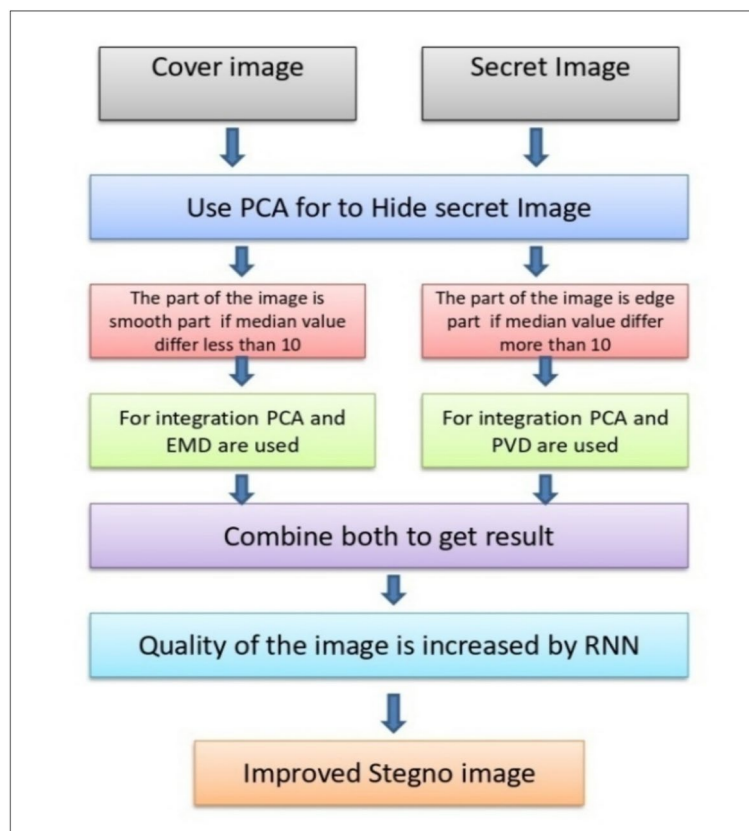


Fig. 3. Flowchart to Embedding image using PCA, EMD and PVD.

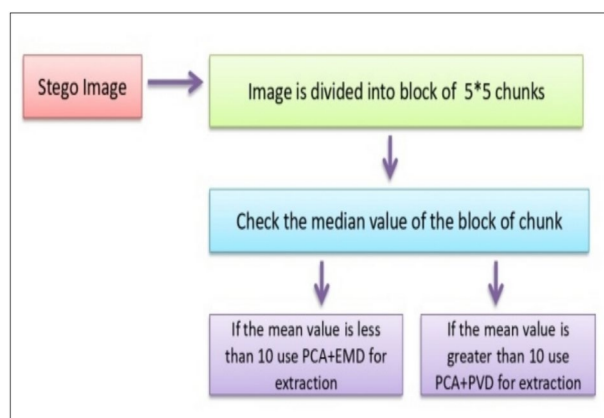


Fig. 4. Flowchart to retrieve image using PCA, EMD and PVD.

Start

Step 1: Choose both the cover image and the hidden image

1.1 The picture is processed in a cross-over manner according to the column wise order and segmented into distinct blocks of dimensions 5×5

1.2 Find out the value of pixels $\text{Diff} = (1/3) \sum_{i=1}^3 |B_c - B_i|$

1.3 If the variation is more than 10, then the insertion technique used is PCA+EMD.

1.4 If the variation is below 10, then the insertion technique PCA combined with PVD is employed.

Step 2: After that, an image of stego will be created; still it will display a lower resolution as displayed in Fig. 2.

Step 3: The technique will be executed for the improved stego picture, as illustrated in Fig. 5.

3.1 The appearance of the stego image created with +EMD and PCA+PVD is not sufficient. As a result, further processing for these images from stego need to be adapted to the specific smart system.

Figure 5 shows a block structure showing the procedure to improve the standard of a stego image with the application of RNN.

3.1 The RNN examines the characteristics that comprise the stego picture and the individual bits of private information. The RNN process provides the altered bits and the characteristics of the fresh stego picture.

The fresh characteristics of stego pictures will eventually be examined in comparison with the features of cover pictures.

3.2 Should the results from this compared prove equivalent, the procedure will collect the modified bits of secret information obtained through the RNN method. An enhanced stego picture can be generated using these modified bits.

3.3 In situations where characteristics lack comparing, the features of the stego photo will need additional processing with RNN with propagation backward their studies.

Step 4: The method for extraction is displayed in Fig. 4

Stop

The procedure of incorporating a hidden image into the cover picture while employing PCA, EMD, PVD, and RNN to enhance the ultimate stego picture is shown in Fig. 3. We use PCA to conceal a hidden picture after inputting the cover and hidden images. The image's evaluation has identified a smoothness area (median value variance of 10) and the edge area (median value difference of 10). PCA and EMD are utilized for calculating smoothness areas, and PCA and PVD are used to connect edge zones. The starting stego picture gets generated by combining the outputs from these various points. An RNN further develops the stego attributes and hidden elements, enhancing their level of detail and ensuring enhanced integration and visual addition. An improved stego picture with improved integration and resolution is the final product.

Figure 4 demonstrates how to extract secret information through of a stego picture. The stego picture has been split through 5×5 sections, and the mean value for every chunk is examined. PCA + EMD is employed for extracting if the chunk's mean value falls lower compared to 10. If the mean value is higher than 10, PCA + PVD is used to find this information out of it. This allows for guarantee that the extracting technique is adjustable dependent upon where every chunk is generated.

PCA, EMD, PVD algorithm

To develop a mathematical foundation for the PCA + EMD (Empirical Mode Decomposition) or PCA + PVD (Pixel Value Differencing) algorithms for image extraction, let's break down the process in general terms: Image extracting occurs by combines with PCA + EMD (Empirical Mode Decomposing) or PCA + PVD (Pixel Values Differencing), Combining the dimension reduction efficiency of PCA using the characteristic extraction or decomposition techniques of EMD or PVD. Provided above is a mathematical examination of these two methodologies: Fig. 5. The process diagram demonstrates how to utilize an RNN to enhance the quality of an image, starting with obtaining and exporting a poor-quality image. We have normalized an image (adjusting pixel counts to a standardized frequency) for ease of use, and we may transform it into grayscale. After initializing the RNN's variables, the neural network transmits the picture and periodically processes it to enhance and perfect its appearance. The repetition process continues until it encounters the maximum variation threshold or converging criteria, such as minimal adjustments to the output. This is presented while creating the ultimate enhanced results to illustrate the improved picture quality; post-processing methods are used to recreate or optimize the final picture.

PCA + EMD for image extraction

Principal Components Analysis (PCA) operates to decrease Dimensional and emphasize the main characteristics of a public image but Empirical Mode Decomposition (EMD) splits into the picture to multiple intrinsic Mode Functions (IMFs).

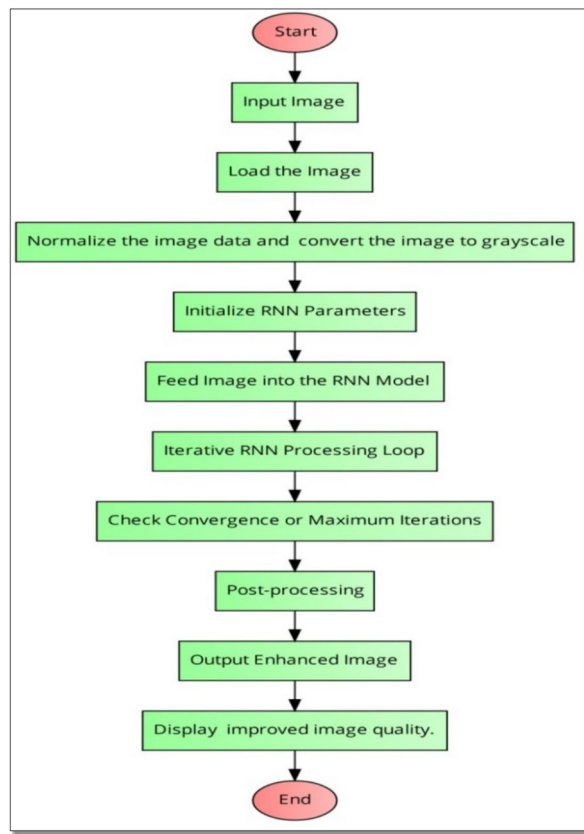


Fig. 5. Steps to retrieve high quality image using RNN.

Step 1: Constructing PCA

Take a the picture matrix X of size $m \times n$:

1. Set up X to have an average of 0.
2. Construct the coefficient matrix:

$$C = \frac{1}{m} X^T X$$

Calculate its eigenvalue λ and eigenvectors V of C . The project the input over the method component to produce $Y = XV$, whereas Y indicates the shortened characteristic collection.

Step 2: Deploying EMD

The PCA- transformed information Y is further divided into a limitless number of Intrinsic Mode Functions (IMFs) using Empirical Mode Decomposition (EMD).

The standard Empirical Mode Decomposition (EMD) approach to calculating an input signal $Y(t)$ includes:

$$y(t) = \sum_{i=1}^N \text{IMF}_{i(t)} + r(t)$$

In which N is the number of internal form operations, and (t) indicates the remainder.

Step 3. Merging of PCA with EMD

Implement PCA to find the main aspects and further breakdown those characteristics utilizing EMD, giving a successful depiction of the attached image.

Step 4. Integrate Fuzzy Logic for Adaptive Feature Selection

Fuzzy Members Parameters are developed to divide characteristics into three categories of significance: low, moderate, and high. This makes it possible for the incorporation of fuzzy reasoning into the process of choosing features. A Fuzzy Rule-Based System (FRBS) is subsequently utilized to make decisions that are adaptive based on image features. More specifically, the architecture keeps parts with low frequencies to keep significant data whenever the contrast between pictures is high and the IMF frequency is low. Minimizing deformations requires removing excessive-frequency IMFs when picture noise is essential. To identify the optimum balance between separating features and reduction of noise, the system continuously combines PCA and EMD in relatively complicated images. This adaptable approach guarantees the characteristics extracted are strong and applicable under various image constraints.

PCA + EMD image embedding approach

Y'_C Is calculated as follows

$$Y'_C + 2^n, \text{ if } df1 > 2^{n-1}, 0 \leq (Y'_C + 2^n) \leq 255$$

$$Y'_C - 2^n, \text{ if } df1 > -2^{n-1}, 0 \leq (Y'_C - 2^n) \leq 255$$

Y'_C otherwise.

Use of PCA + PVD for image extraction

Principal Component Analysis (PCA) and Pixel Value Differene (PVD) for Image Extraction: Pixel Value Differencing (PVD) improves finding features by analyzing changes in the value of pixels, while Principal Component Analysis (PCA) helps in reducing the dimensionality.

Step 1: constructing Physical Vapor Deposition (PVD)

For each pixel pairing (p, q) :

To find the gap:

$$d = |p - q|$$

Separate each pair of dots into groupings according to their variation d . may add or take out data from these sections.

Step 2: Deploying PCA

Collect the changed value of pixels post-PVD using the data matrix X and apply PCA according to the previously described to reduce complexity and identify major attributes:

$$Y = XV$$

Step 3: Merging PCA and PVD:

PVD finds significant differences between pixels pairs, while PCA collects and highlights the most important features from the modified image matrix, allowing rapidly picture retrieval.

The median pixel value difference for every block is

$$Diff = \left(\frac{1}{n}\right) \sum_{k=1}^3 |B_C - B_i|$$

Figure 6a shows the 3×3 covered block structure is an important building block for analyzing and changing digital images. Many situations can benefit from its use, particularly in determining the composition of materials and the extent of surface wear. Using a distinctive grayscale or color intensity value, every pixel (P1–P9) indicates distinct regions that comprise the picture. Because of the way the space is set up, computational tasks like finding surface defects, figuring out porosity, and highlighting worn features can all be done separately. The structured layout makes it easier to use techniques for getting or measuring features, like finding edges for figuring out where paths or patterns for evaluating micro structural integrity. Including covering blocks in your study lets you perform a systematic investigation of their surface integrity, which in turn enables you to develop insight on the material's wear performance or top features. Figure 6b shows detailed pixels of the steno block.

Step 4: Integrate Fuzzy Logic for Adaptive Feature Selection

Low Information Significance, Average Feature Significance, and Strong Feature Significance are the three groups into that Fuzzy Membership Functions are generated to define the significance of features, and this in turn improves the flexibility of the extraction of features. The next thing to do is to maximize the photo the extraction procedure utilizing a Fuzzy Rule-Based System (FRBS) that taking taken into consideration different image features. For pictures with significant contrast and small pixel distinction the procedure will carefully apply PVD to the corners of the image to be able to keep important details. After applying PVD, noise filtering is done avoid distortions in images with high levels of noise. For images with a moderate level of complexity, the PCA choice of components is made on the fly to maximise feature retention and reduce data duplication. By incorporating fuzzy logic, we can make our image extraction process smarter and more flexible, making it more resilient to varying image conditions.

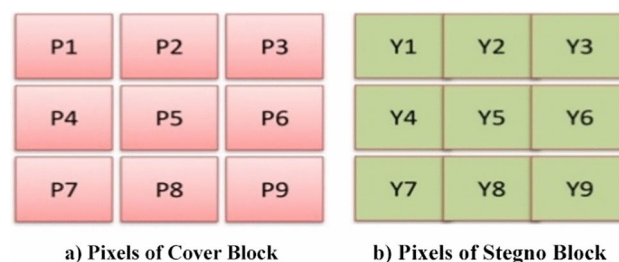


Fig. 6. (a) Pixels of cover block. (b) Pixels of Stegno Block.

The PCA + PVD image embedding approach

Y'_C Is calculated as follows

$$Y''_C, \text{ if } df1 > 2^{n-1}, 0 \leq Y''_C \leq 255$$

$$Y'_C - 2^n, \text{ if } df1 > -2^{n-1}, 0 \leq (Y''_C - 2^n) \leq 255$$

Y''_C Otherwise.

For the computation of Y''_C , the value can be determined by verifying all three criteria with the measure $df1$ as well as a range restriction. It provides the value of Y''_C if $df1$ is higher than 2^{n-1} and Y''_C falls between 0 and 255. In contrast, if $df1$ exceeds than 2^{n-1} and the modified number $(Y''_C - 2^n)$ falls somewhere between 0 and 255, then Y''_C will be calculated as $Y'_C - 2^n$, resulting in the same idea as a modular rearrangement. Y''_C Always changes to Y''_C in all other circumstances. This method of carrying out things making sure that the value Y''_C stays between the 0 and 255, something that's essential for activities like digitized encoded data, where data need to be translated into likely bit stages, ranging like eight-bit color or signal conditioning structures.

Recurrent neural network

RNNs have a comparable method for getting data in and getting it back as any other kind of deep neural layout. However the processes by which it moves from input to output remain the same. We have different matrices of weight for every dense system in deep neural networks, but the amount of weight throughout the entire RNN system keeps the exact same. It finds out a kind of concealed operation.

Figure 7 demonstrates how to reduce size to make the model of neural networks simple to follow. The picture depicts an ordinary multi-layer neural network structure. It has input, hidden, and layers for output. The architecture of a network has been broken down into three components: the input (x), hidden (h), and output (y) layers. The result happens by transforming the structure in the opposite direction and compressing it. The summary of the links between the layers, the labels A, B, and C, shows the simplified connections among the input, hidden, and output stages. Recurrent Neural Network (RNN) is designed to manage segments data sets via connects which proceed back and forth between one another. It keeps knowledge live continuously. the formulas for an elementary RNN:

The base RNN calculations

Hidden state

$$h_t = f(W_h x_t + U_h h_{t-1} + b_h).$$

h_t : Hidden state at moment t , x_t , the subsequent state was concealed: Which was entered in at the time t , h_{t-1} : State which had been concealed in the final step of time; W_h : A collection of weights over input to the concealed layer, U_h : The weight matrices for a layer that moves via hidden to secret (recurrent link); b_h : The value of the bias vector, f : the activating function typically has the form $\tanh$ or ReLU .

Output Calculation:

$$y_t = g(W_y h_t + b_y)$$

y_t : The result at time t ; W_y : Dimensions matrices that is secret final phase; b_y : This serves as the preference vector for the result of the layer; g : Result for the function of activation.

It is achievable to utilize an RNN for modeling sequences since it has a hidden state h_t which retains information to every one of stages until t .

The state that is hidden changes at every interval t according to the present input X_t as well as the secret state to the time phase prior to that, h_{t-1} .

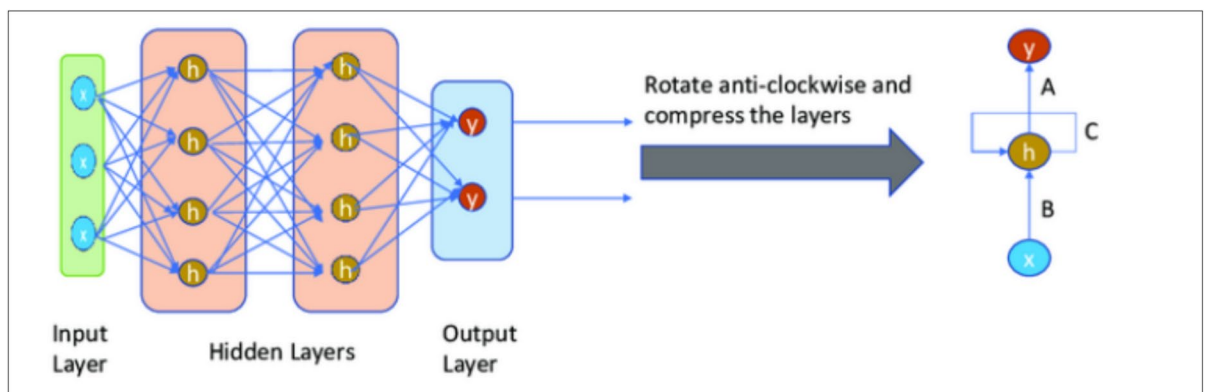


Fig. 7. Structure of recurrent network²⁴.



Fig. 8. Cover images i) Zebra ii) Elephant iii) Lion iv) Deer.



Fig. 9. Secret images used i) Waterfalls ii) Mountain iii) Sea²⁷.



Fig. 10. Resulting stegno image of PCA-PVD-EMD.

The current secret state h_t is applied for figuring out the result y_t

Result and discussion

The suggested approach was employed in directing our investigations that were subsequently verified with the picture collection mentioned below it. Figure 8 shows four various cover illustrations which could be utilized for testing the recommended way. It depicts a number of cover photos, such as Zebra, Elephant, Lion, and Deer. As demonstrated through Fig. 9, every one of these cover pictures is evaluated toward three separate hidden images of varying proportions. In this experiment, all three secret images of Waterfalls, Mountain, and Barabara are used. The images are 5 kb, 10 kb, and 15 kb in capacity. The findings of this research are shown with a number of pictures.. The images of stego produced via the Vernam algorithm as well as the outcomes of PCA replacement with PVD and EMD are depicted in Figs. 11 and 12, respectively. The results of the proposed technique are illustrated in Fig. 10.

The three different hidden pictures are examined at with each cover image. The images from stego which were created following PCA-PVD-EMD are displayed in Fig. 10, and the pictures that had been formed by the Vernam algorithm have been demonstrated in Fig. 11. The Vernam cipher (one-time pad) produces a ciphertext that appears as random noise when visualized because every plaintext bit is XORed with a truly random key bit. This results in an output that lacks discernible patterns, making it theoretically unbreakable if the key is as long as the message and used only once. Any attempt to visualize the ciphertext would show a uniform distribution of bits, resembling static or white noise.



Fig. 11. Resulting Image of Vernam algorithm.



Fig. 12. The resulting image of improved quality stegno algorithm 1.

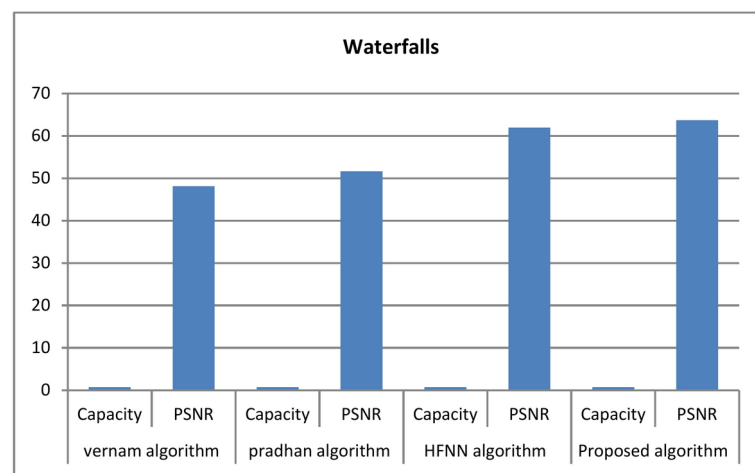


Fig. 13. PSNR values for different payload capacity of Waterfalls image.

The stego image via Fig. 12 was created through an RNN on PCA replacement through PVD and EMD. During the extraction stage, the inserting cycle is performed backward, requesting to extricate the scrambled hidden picture from the stego picture. It does not need the cover picture. Figure 13 depicts the extracted covered-up image and stego image sections. The example shows how the suggested approach could be adopted to successfully distinguish the hiding picture without first changing the picture's character. Our unique approach has been evaluated with a number of criteria, including capacity, picture fidelity, KL differences, MSE, and PSNR. The K-L difference computation is used for safety reasons. It illustrates the efficiency of the steganography method. The following equation demonstrates the formula for KL difference.

$$KL(P||Q) = -\sum_{\{x \in X\}} P(x) \cdot \log \left(\frac{\{Q(x)\}}{\{P(x)\}} \right)$$

The proposed methodology is contradicted of traditional methodologies in Table 1. In this, it has implemented the current process offered by Pradhan et al.²⁹, which is using PVD and EMD in combination with LSB replacement. Based upon the principle, Yaseen et al.³⁰ include an use as. I demonstrate in the table that the

recommended method achieves the present that with a as well as a comfortable margin lesser mean square error. Lion's image appears as a cover picture from the first as well as the two columns of Table 1, together with images of Sea, Waterfalls, aircraft.

Table 2 examines the way the four distinct algorithms, Vernam, Pradhan, HFNN, and a proposed algorithm do in two essential indicators: The capacity and image Fusion, in a range of covering and secret picture pairs (like Lion, Elephant, Waterfalls, and Mountain). The capability data stay constant throughout each algorithm at amounts like 0.72, 1.6, 2.3, etc., however the ****Image Fusion**** metric shows significant variances. The proposed algorithm works far better than others since it receives the greatest integration values (like 0.99587, 0.99789, and 0.99987), indicating that that it produces greater quality pictures and performs better. HFNN also does properly, although its combining values are a little lesser. Vernam and Pradhan come in second and third, respectively, with results that are lower in general, particularly in difficult circumstances. As a whole, the proposed method does a more effective job of merging images than any of the other techniques that were examined at, therefore ranking as the most successful one. There are some obvious distinctions among the proposed and current approaches. Table 2 additionally examines two other parameters such as: picture fidelity and capacity. Table 2 illustrates how the recommended procedures enhances image quality.

Figure 13 demonstrates the way well the Vernam, Pradhan, HFNN, as well as Developed algorithms works for Capacity and PSNR. All of the methods have concerning identical Capacity, however the suggested technique has a greater PSNR, indicating it offers the best quality of picture. It gets followed by HFNN, Pradhan, and Vernam. It demonstrates that this proposed approach performs more efficiently than the other ones.

Figures 14 and 15 shows all algorithms display identical capacity levels for both the Sea and Mountain images; however, the PSNR (image resolution) continues to improve from Vernam to Proposed. The suggested method gives the highest PSNR in both instances. HFNN, Pradhan, and Vernam follow closely behind. In both scenarios, the suggested method provides better results in terms of preserving the images' clarity.

The three pictures (Figs. 16, 17 and 18) all show the same measurement range for the Waterfalls, the plane, and Sea. However, the average square error (MSE) is very different. The Vernam method consistently has the greatest MSE, which means that the image quality is even worse. Pradhan and HFNN follow it. On the other hand, the recommended technique includes the least mean square error (MSE), indicating that it performs most effectively. The suggested method preserves picture quality more accurately and produces fewer distortions across each picture. The suggested method is most effective at decreasing defects since the pattern continues unchanged regardless of the image that is utilized.

Analysis of time complexity

MATLAB, which 2020 was utilized on a system running Microsoft Windows 10 and featuring an i7 8th gen chip to determine the overall time required for processing. Using the biggest packet size, Table 3 demonstrates the time the suggested approach needs to run comparing with other techniques such as the Vernam algorithm, LSB + PVD + EMD, HoG + PVD + LSB. The importing and extraction intervals demonstrate the time it takes on average to operate via seven images in grayscale. The approach is unsuccessful when compared PCA + EMD or PCA + PVD since it include an enhancement to quality module. Through a total execution time of 3.6 s, the PCA + EMD + PVD technique is the quickest. DT-CWT + KNN, on the opposite hand, and needs long since it utilizes more complex techniques and the KNN to categorize chunks. This approach needs 5 s, while the total Vernam technique uses 5.1 s. It requires lengthier for the recommended method to be functional compared to PCA + PVD + EMD, however it provides more reliability, protection, and payload weight compared to all the others.

Conclusion

In this article, we outline an innovative steganography as method that combines reverse propagation gaining knowledge, artificial neural networks, and fuzzy reasoning. This approach combines PCA as well as EMD strategies to enhance privacy, load capability, and image resolution for steganographic techniques through integrating the encryption of images via LSB substitution. The suggested approach, that employs a recurrent neural network (RNN), gives up novel opportunities for making stenography work smoother. The technique improves processes better through the combination of the encryption of images with LSB replacement through PVD as well as EMD. It do an extensive evaluation of two present techniques using various parameters such as PSNR, MSE, the image Fidelity, and the payload capacity, in addition to multiple covering and hidden images (5 KB, 10 KB, and 15 KB). The findings indicate that this suggested method performs due to PSNR values are now greater, reliability is more, and the mean square error is reduced. This research additionally addresses about just how durable it is to combine image encryption, LSB substitution, PVD, and EMD algorithms. The combination of Recurrent Neural Network additionally makes guarantee that the quality of the stego picture is enhanced. The researchers can enhance the proposed approach even further in the future by adding machine learning as well as techniques for optimization. They require making it more secure, more effective at stego pictures, and able to handle additional information. This novel approach not only enhances security but also ensures efficient data embedding with minimal distortion. By leveraging advanced neural networks and optimization techniques, the method strengthens steganographic robustness. Future advancements could explore deep learning integration for adaptive steganography and improved resilience against attacks.

Cover image		Lion			Elephant			Zebra			Lincoln			Deer		
Hidden image	Capacity	Waterfalls	Mountain	Sea	Waterfalls	Mountain	Sea	Waterfalls	Mountain	Sea	Waterfalls	Mountain	Sea	Waterfalls	Mountain	Sea
	Image fusion	0.72	1.6	2.3	0.96159	0.96047	0.95954	0.89953	0.8988	0.897695	0.96165	0.96097	0.95999	0.95499	0.95413	0.95313
Vernam algorithm	Capacity	0.72	1.6	2.3	0.72	1.6	2.3	0.72	1.6	2.3	0.72	1.6	2.3	0.91	2.02	2.92
	Image fusion	0.75972	0.75872	0.75772	0.76807	0.76707	0.76607	0.86278	0.86178	0.860775	0.5697	0.5687	0.5677	0.91671	0.91571	0.91471
HFNN algorithm	Capacity	0.72	1.6	2.3	0.72	1.6	2.3	0.72	1.6	2.3	0.72	1.6	2.3	0.91	2.02	2.92
	Image fusion	0.98597	0.98497	0.98397	0.99083	0.98983	0.98883	0.97901	0.97801	0.977014	0.88561	0.88461	0.88361	0.96608	0.96508	0.96408
Proposed algorithm	Capacity	0.72	1.6	2.3	0.72	1.6	2.3	0.72	1.6	2.3	0.72	1.6	2.3	0.91	2.02	2.92
	Image fusion	0.99587	0.98678	0.99671	0.99789	0.99987	0.99698	0.98914	0.98793	0.983454	0.89765	0.89567	0.89891	0.97658	0.9789	0.98908

Table 2. Performance of Stego images based on image fusion and capacity.

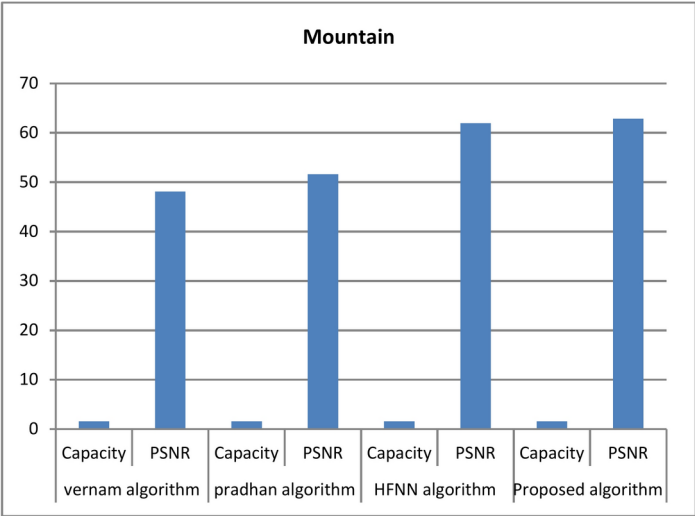


Fig. 14. PSNR values for different payload capacity of Mountain image.

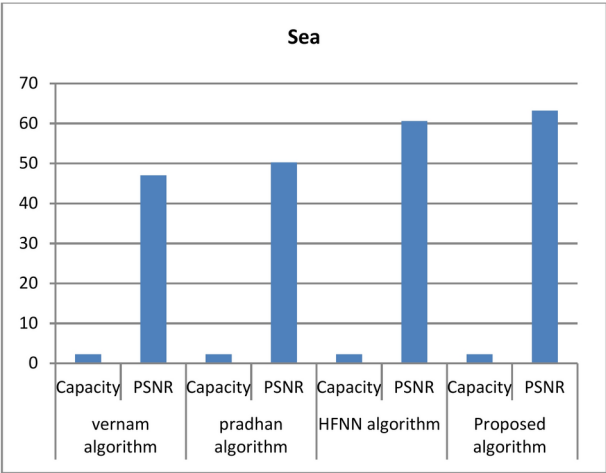


Fig. 15. PSNR values for different payload capacity of Sea image.

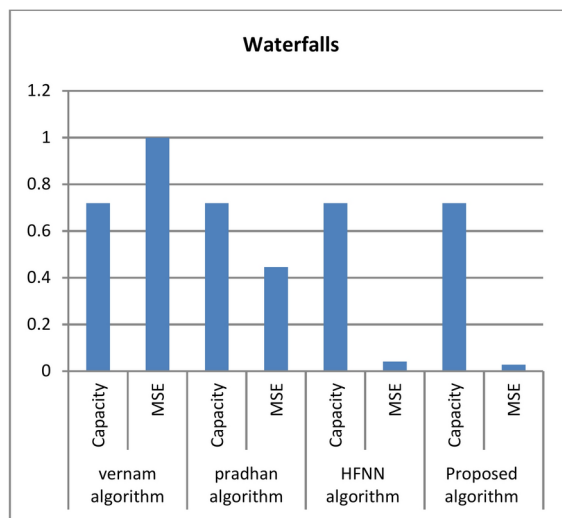


Fig. 16. MSEs for different payload capacities for Waterfalls image.

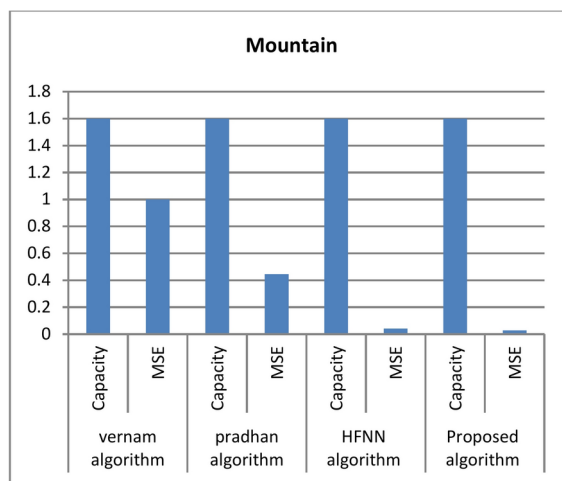


Fig. 17. MSEs for different payload capacities for Mountain image.

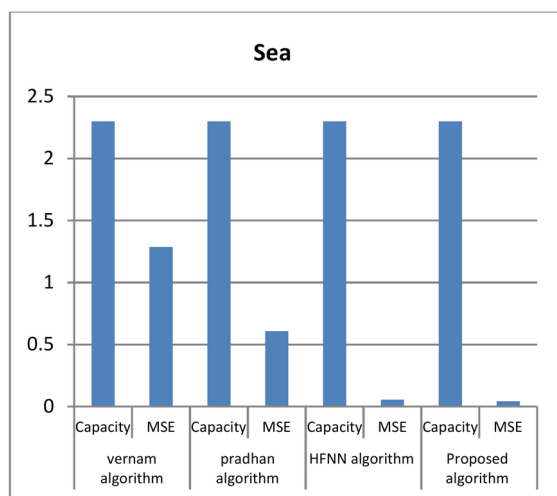


Fig. 18. MSEs for different payload capacities for Mountain image.

Methods	Embedding time (s)	Extraction time (s)	Total time (s)
Vernam algorithm	2.6	2.7	5.3
LSB + PVD + EMD	2.3	1.6	3.9
DT-CWT + KNN	3.5	3.8	7.3
HFNN	2.9	2.1	5
Proposed	2.7	2	4.7

Table 3. Time complexity analysis.

Data availability

The datasets used and/or analyzed during the current study available from the corresponding author on reasonable request.

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Author contributions

R. Kanimozhi: Problem Selection, Algorithm, implementation, Results, coding, testing V. Padmavathi: Experimental Results, Algorithm part and Editing.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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